

Improving Retrieval-Augmented Feedback for LLM-based Dialogue Coaching Systems

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Abstract

Large language models are increasingly used as dialogue coaching assistants for tasks such as interview preparation and elevator pitch practice, but standalone LLM feedback is often generic and weakly tied to the user's actual response. We investigate whether retrieval-augmented generation can improve LLM-based dialogue coaching feedback, and whether tag-aware retrieval offers additional benefits over text-similarity-based retrieval. We design a compact taxonomy of coaching tags describing communication strengths and weaknesses, and use these tags to filter retrieved documents from a corpus of dialogue examples, feedback examples, and guidelines. We compare five system variants spanning no retrieval, sparse and dense retrieval, and their tag-filtered counterparts, using an LLM-as-judge evaluation across relevance, specificity, actionability, helpfulness, and groundedness. Retrieval most consistently improves groundedness, with all retrieval-based variants outperforming the no-retrieval baseline. We find no clear advantage of dense retrieval over BM25 in our small-corpus setting. Tag-aware retrieval rarely dominates any single dimension but is the best overall method on 68.8% of inputs, compared with 18.8% for the no-retrieval baseline. These results suggest that for subjective, pedagogical generation tasks, the relevance of retrieved documents matters more than the choice of retrieval algorithm, and that an intermediate tag representation usefully bridges user diagnosis and document selection. Our code is available at <https://github.com/grace-go/cs510-dialogue-coaching>.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Natural language generation*.

Keywords

retrieval-augmented generation, large language models, dialogue coaching, feedback generation, tag-filtered retrieval, dense retrieval, LLM-as-judge evaluation

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1 Introduction

Large language models (LLMs) are increasingly used as dialogue coaching assistants for tasks such as job interview preparation, elevator pitch practice, presentation rehearsal, and professional communication training. These systems are attractive because they can provide immediate natural-language feedback without requiring a human coach. For example, a user can submit a transcript of an interview answer and ask the model to identify weaknesses, suggest improvements, and rewrite the response. However, despite their fluency, standalone LLM-based coaching systems often produce feedback that is overly generic, insufficiently grounded in specific coaching criteria, and weakly connected to the user's actual response. Feedback such as "be more specific," "improve structure," or "sound more confident" may be correct, but it is often not actionable enough to help users understand exactly what to change.

This project studies how retrieval-augmented generation (RAG) can improve LLM-based dialogue coaching. In a RAG framework, a system retrieves relevant external information before generating a response. RAG has been widely explored for tasks such as question answering, factual grounding, and document-based generation. In those settings, retrieval is mainly used to provide facts or evidence that the language model may not know internally. In contrast, our work applies retrieval to a more subjective and pedagogical task: generating useful feedback for dialogue performance. The retrieved information in our system is not factual knowledge in the traditional sense, but coaching knowledge, including feedback examples, strong and weak dialogue examples, and communication guidelines.

The central research question of this project is: *Can tag-filtered retrieval improve the specificity, relevance, and actionability of LLM-generated feedback for dialogue coaching tasks?* More specifically, we compare a standalone LLM feedback generator with RAG-based variants that retrieve coaching context using sparse retrieval, dense retrieval, and tag-filtered retrieval. The main intuition is that feedback quality can be improved when the generator is conditioned on examples and guidelines that match the user's specific strengths and weaknesses. For instance, if a user's elevator pitch is vague and lacks a clear career goal, the system should retrieve feedback examples and guidelines related to tags such as `too_general`, `missing_details`, and `wordy`, rather than retrieving only semantically similar transcripts.

The novelty of this project is the use of *tag-aware retrieval filtering* for dialogue coaching feedback. Instead of retrieving examples only by surface-level text similarity, our system first represents both user inputs and corpus items using a controlled set of coaching tags. These tags describe broad strengths and weaknesses, such as `clear_goal`, `specific_example`, `confident_tone`, `too_general`, `poor_structure`, and `weak_closing`. The retrieval module then uses these tags to prioritize coaching materials that are relevant to the user’s actual communication problem. This design is important because two dialogue responses can be textually similar while requiring different feedback, and two responses can be textually different while sharing the same underlying weakness. For example, an interview answer and an elevator pitch may contain different words, but both may suffer from missing details or unclear goals. A tag-filtered approach allows the system to retrieve based on coaching needs rather than only lexical or semantic similarity.

This novelty matters because dialogue coaching is not only a text-matching problem. Effective feedback depends on diagnosing communicative weaknesses and selecting appropriate guidance. A retrieval system that only retrieves similar text may fail to retrieve examples that target the right improvement dimension. By introducing tags as an intermediate representation, our system makes retrieval more interpretable and controllable. The tags also make it easier to analyze why a particular document was retrieved, which weaknesses the system focused on, and whether the generated feedback addressed the user’s needs.

In this project, we implement a prototype RAG-based dialogue coaching system with five main variants: no retrieval, sparse retrieval using BM25, dense retrieval using sentence embeddings, tag-filtered retrieval using BM25, and tag-filtered dense retrieval. The system takes a user transcript as input, predicts or uses assigned tags, retrieves relevant coaching context from a corpus, and generates structured feedback using an LLM. The feedback includes an overall assessment, strengths, areas for improvement, specific revision suggestions, and an improved version of the user’s response. We evaluate the system by comparing the generated feedback across variants, focusing on relevance, specificity, actionability, usefulness, and groundedness.

The contributions of this work are as follows. First, we formulate dialogue coaching feedback as a retrieval-augmented generation problem. Second, we design a compact tag taxonomy for representing coaching strengths and weaknesses. Third, we implement a tag-filtered retrieval pipeline that combines coaching tags with sparse and dense retrieval methods. Fourth, we provide an experimental framework for comparing standalone LLM feedback with retrieval-augmented feedback. Together, these contributions show how information retrieval techniques can be adapted to improve LLM-based communication coaching systems.

2 Related Work

Recent work on AI-assisted communication coaching has explored how large language models and conversational systems can support interview preparation, public speaking practice, and workplace communication training. At the same time, retrieval-augmented generation (RAG) has become an important framework for grounding

language model outputs using external information. Our work connects these two research directions by applying retrieval-augmented generation to dialogue coaching feedback.

Several prior studies have investigated AI systems for interview and communication training. [3] introduces an interactive interview coaching system that supports reflective learning through simulated dialogue and AI-generated feedback. The work demonstrates that conversational feedback systems can help users practice interviews and reflect on their communication strategies. Similarly, [9] studies how professionals perceive AI-based communication coaching systems in workplace settings. The paper highlights the growing demand for scalable communication coaching tools while also identifying concerns about feedback quality, trust, and usefulness. In addition, [4] examines expert opinions on AI-generated speaking feedback and emphasizes the importance of actionable, context-aware coaching rather than generic advice. These works motivate the importance of AI-assisted dialogue coaching, but they primarily focus on interaction design, user perception, or coaching effectiveness rather than retrieval-based feedback generation.

Meanwhile, retrieval-augmented generation has emerged as a major approach for improving large language model outputs. The foundational RAG framework proposed in [7] combines a retriever and a generator so that external documents can be retrieved and incorporated during generation. RAG has been widely used for question answering, factual grounding, and knowledge-intensive NLP tasks because retrieval can improve grounding and reduce hallucination. More recent survey papers, including [5] and [6], provide comprehensive overviews of sparse retrieval, dense retrieval, retrieval optimization, and retrieval-grounded generation architectures. These surveys discuss important challenges such as retrieval quality, noisy retrieved context, prompt construction, and retrieval evaluation. Our work builds on this literature by applying sparse retrieval, dense retrieval, and retrieval filtering techniques to coaching-oriented feedback generation rather than factual knowledge retrieval.

Prior research has also explored retrieval-enhanced dialogue generation systems. [8] proposes a dialogue generation framework that first retrieves related examples and then refines them into a final response. The paper argues that standalone neural generators often produce generic and repetitive outputs, while retrieved examples can improve specificity and diversity. Similarly, [1] studies modular dialogue generation pipelines that retrieve external knowledge before generating responses. These works are closely related to our approach because they demonstrate how retrieval can improve dialogue generation quality. However, their focus is primarily on generating conversational responses rather than generating evaluative coaching feedback.

Finally, recent work on AI-generated feedback systems further motivates our approach. [2] investigates systems that generate critiques, revisions, and iterative feedback for educational tasks. The paper highlights the importance of actionable and revision-oriented feedback, which aligns with our use of revision-pair coaching examples. While these systems focus on educational feedback generation, our work specifically targets communication coaching tasks such as interviews, presentations, and elevator pitches.

Compared to prior work, our project differs in two important ways. First, we formulate dialogue coaching feedback as a retrieval-augmented generation problem rather than only a conversational simulation or dialogue generation problem. Second, our system introduces tag-filtered retrieval for coaching feedback. Instead of retrieving documents solely based on lexical or semantic similarity, our retrieval pipeline incorporates interpretable coaching tags representing strengths and weaknesses such as `too_general`, `missing_details`, `clear_goal`, and `poor_structure`. This allows the system to retrieve coaching materials based on communicative needs rather than only textual similarity. Through this design, our work combines ideas from retrieval-augmented generation, dialogue systems, and AI communication coaching to investigate whether tag-aware retrieval can improve the specificity, relevance, and actionability of LLM-generated coaching feedback.

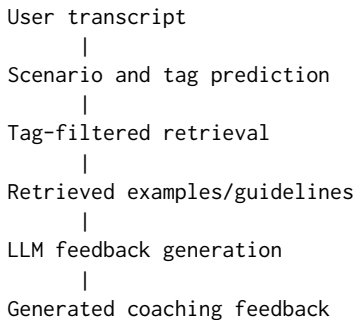
3 Methodology

3.1 System Overview

We design a retrieval-augmented dialogue coaching system that generates feedback for user-submitted dialogue transcripts. The system supports coaching scenarios such as elevator pitches and behavioral interview answers. Given a transcript, the system identifies relevant coaching tags, retrieves useful coaching context from a corpus, and prompts a large language model (LLM) to generate structured feedback. The system is designed not only as a prototype coaching tool, but also as an experimental framework for comparing different retrieval strategies.

The overall pipeline consists of five main components: input processing, tag prediction, retrieval, feedback generation, and evaluation. First, the system receives a user transcript and its scenario type. Second, the tag prediction module assigns a small set of coaching tags that describe the strengths and weaknesses of the transcript. Third, the retrieval module searches a corpus of coaching examples and guidelines. Fourth, the retrieved context is passed to an LLM, which generates feedback. Finally, the generated feedback from different system variants is compared using automatic evaluation criteria.

Conceptually, the system follows the architecture below:



This modular design allows us to isolate the effect of retrieval on feedback quality. The same user inputs and generation prompt are used across system variants, while the retrieval method is changed. This makes it possible to compare whether retrieval improves feedback and whether tag-filtered retrieval provides benefits beyond standard sparse or dense retrieval.

3.2 Corpus Design

The retrieval corpus consists of coaching-related documents. The corpus is stored as a JSON file containing a list of document objects. Each document contains a unique identifier, document type, scenario, text content, and tags. We include three types of corpus items: `revision_example`, `feedback_example`, and `guideline`. A `revision_example` contains a weak or incomplete user response paired with an improved version that demonstrates how the response can be revised. A `feedback_example` contains feedback written for a sample response, and a `guideline` contains general communication principles.

The following example shows the format of a corpus item:

```

{
  "id": "feedback_001",
  "type": "feedback_example",
  "scenario": "elevator_pitch",
  "text": "Your pitch is understandable, but it is
too general.
Add one specific project, explain what role you are
looking for, and end with a clear next step.",
  "tags": ["too_general", "missing_details", "weak_closing"]
}

```

The full corpus file has the following structure:

```

[
  {
    "id": "feedback_001",
    "type": "feedback_example",
    "scenario": "elevator_pitch",
    "text": "...",
    "tags": ["too_general", "missing_details"]
  },
  {
    "id": "guideline_001",
    "type": "guideline",
    "scenario": "interview",
    "text": "...",
    "tags": ["logical_structure", "specific_example"]
  }
]

```

We chose JSON as the corpus format because it is simple, human-readable, and easy to process in Python. Storing the corpus as a single JSON array also makes it easy to inspect the full dataset, edit examples manually, and load all documents at once during experimentation. Since the goal of this project is to evaluate retrieval strategies rather than build a large-scale benchmark dataset, a manually constructed and interpretable JSON corpus is appropriate for the first version of the system.

This corpus structure was chosen for three reasons. First, it is interpretable: each retrieved document can be inspected by its text, scenario, type, and tags. Second, it supports multiple retrieval sources. Revision examples, feedback examples, and guidelines provide different kinds of coaching information. Revision examples demonstrate how weak or incomplete responses can be improved, feedback examples show how critique may be phrased, and guidelines provide general principles that can guide the generation model.

Third, the structure is extensible. New scenarios, tags, and document types can be added without changing the overall pipeline.

3.3 Tag Taxonomy

A key design choice in our system is the use of a controlled tag taxonomy. Each user input and corpus document is associated with coaching tags that describe communication strengths and weaknesses. We use a compact tag set instead of a large fine-grained taxonomy to reduce annotation difficulty and improve retrieval reliability.

The strength tags are:

```
clear_goal, logical_structure, conciseness,
specific_example, relevant_experience,
confident_tone
```

The weakness tags are:

```
too_general, missing_details,
poor_structure, uncertain_tone,
vague_example, weak_closing
```

These tags were selected because they are broad enough to apply across different coaching scenarios while still being specific enough to guide retrieval and feedback generation. For example, `too_general` and `missing_details` are common weaknesses in both elevator pitches and interview answers. Similarly, `specific_example` and `logical_structure` are important qualities in many types of professional communication.

We initially considered using a larger set of tags, including more detailed labels such as `quantified_impact`, `missing_STAR`, `weak_result`, and `low_confidence`. However, a larger tag space creates several challenges. It makes annotation less consistent, increases the chance that relevant documents have non-overlapping tags, and makes retrieval more sparse. Therefore, we intentionally use a smaller set of general tags. This choice reflects a trade-off between expressiveness and robustness. For this project, we prioritize reliable retrieval and clear analysis over fine-grained diagnosis.

3.4 Tag Prediction

The tag prediction module assigns coaching tags to the user's transcript. In the current prototype, tags can be assigned manually or predicted using a simple rule-based function. Manual tagging is useful for controlled experiments because it reduces noise and allows us to focus on the effect of retrieval. A fully automated system could later replace this component with an LLM-based tagger or a supervised classifier.

The tag prediction function has the following interface:

```
predict_tags(user_input: str) -> List[str]
```

For example, given the input:

```
Hi, I am a computer science student. I like AI and I
worked on some projects. I want to get an internship.
```

the tagger may assign the following weakness tags:

```
too_general
missing_details
vague_example
weak_closing
```

These tags indicate that the response lacks concrete details, does not include a specific project or achievement, and does not end with a strong closing statement. The retrieval module can then prioritize corpus items that address these weaknesses.

The main challenge in tag prediction is that dialogue quality is subjective. A single transcript can have multiple possible interpretations, and different annotators may assign different tags. We address this challenge by using broad and interpretable tags. This reduces ambiguity and makes the tags easier to apply consistently.

3.5 Retrieval Methods

We implement and compare three retrieval approaches: sparse retrieval, dense retrieval, and tag-filtered retrieval. The purpose of including multiple retrieval methods is to determine whether retrieval helps feedback generation and whether tag-aware retrieval improves over standard text-similarity-based retrieval.

3.5.1 Sparse Retrieval with BM25. Sparse retrieval is implemented using BM25. BM25 ranks corpus documents according to lexical overlap with the user transcript. Each document is tokenized, and the BM25 score is computed between the input transcript and each corpus item. The top-ranked documents are returned as retrieved context.

The sparse retrieval process can be summarized as follows:

```
Input transcript
|
Tokenize transcript and corpus documents
|
Compute BM25 scores
|
Return top-k documents
```

BM25 is included because it is a strong and interpretable traditional information retrieval baseline. It is effective when relevant documents share important keywords with the query. However, BM25 has limitations for dialogue coaching. A user response may require feedback about specificity or structure even if the transcript does not contain those exact words. Therefore, BM25 may miss useful coaching guidelines when lexical overlap is low.

3.5.2 Dense Retrieval with Sentence Embeddings. Dense retrieval uses sentence embeddings to represent the user transcript and corpus documents as vectors. We use a SentenceTransformers model to encode each text, then compute cosine similarity between the user transcript and each corpus item. The highest-scoring documents are returned as retrieved context.

The dense retrieval process is:

```
Input transcript
|
Encode transcript and corpus documents
|
Compute cosine similarity
|
Return top-k documents
```

Dense retrieval is included because it can capture semantic similarity beyond exact word overlap. For example, a transcript saying "I worked on some projects" may be semantically related to a guideline about adding concrete evidence, even if the exact words differ.

However, dense retrieval can still retrieve documents that are semantically similar but not pedagogically useful. This limitation motivates our tag-filtered retrieval method.

3.5.3 Tag-Filtered Retrieval. The main retrieval method in this project is tag-filtered retrieval. Instead of ranking all corpus documents only by text similarity, the system first uses coaching tags to identify documents that are relevant to the user’s strengths or weaknesses. This allows retrieval to focus on the user’s communication needs rather than only surface-level similarity.

Given a set of predicted tags for the input and a set of tags for each corpus item, we compute tag overlap as:

$$\text{tag_overlap}(q, d) = |T_q \cap T_d|$$

where T_q is the set of tags assigned to the user query and T_d is the set of tags assigned to a corpus document. A simple version of tag filtering keeps documents with at least one overlapping tag:

```
filtered_docs = [
    doc for doc in corpus
    if len(set(predicted_tags) & set(doc["tags"])) > 0
]
```

After this filtering step, we apply BM25 or dense retrieval within the filtered subset. This gives two tag-aware variants: tag-filtered BM25 and tag-filtered dense retrieval.

We also consider a soft scoring version in which tag overlap is added to the retrieval score:

$$\text{final_score}(q, d) = \text{retrieval_score}(q, d) + \lambda \cdot \text{tag_overlap}(q, d)$$

where λ controls the importance of tag overlap. This soft version is useful because strict filtering can be too aggressive when the corpus is small or when tag prediction is imperfect. By adding tag overlap as a ranking signal, the system can prefer documents with matching coaching needs while still preserving documents that are semantically relevant.

Tag-filtered retrieval is the core novelty of our method. The motivation is that dialogue coaching requires diagnosing the user’s communication problem. Two responses can be textually different but share the same weakness, such as being too general or lacking a strong closing. Similarly, two responses can be textually similar but require different feedback. Tags provide an intermediate representation that connects user diagnosis with retrieval.

3.6 Feedback Generation

After retrieval, the system constructs a prompt for the LLM. The prompt includes the user transcript, scenario type, predicted tags, retrieved documents, and instructions for generating structured feedback. The generated feedback follows the same format for all system variants:

1. Overall assessment
2. Strengths
3. Areas to improve
4. Specific revision suggestions
5. Improved version

We use the llama-3.1-8b-instant model through the Groq API as the primary large language model for feedback generation.

This model was selected because it provides fast inference speed and relatively low computational cost while still producing high-quality conversational outputs. Since our experiments require generating feedback across multiple retrieval variants and evaluation settings, efficient inference is important for reducing latency and API usage.

A simplified prompt template is shown below:

You are a dialogue coaching assistant.

Scenario:
{scenario}

User transcript:
{transcript}

Detected tags:
{predicted_tags}

Retrieved coaching context:
{retrieved_documents}

Generate feedback with:

1. Overall assessment
2. Strengths
3. Areas to improve
4. Specific revision suggestions
5. Improved version

Rules:

- Be specific.
- Refer directly to the user's transcript.
- Use the retrieved context when relevant.
- Give actionable suggestions.

We use a fixed feedback structure to make outputs easier to compare across system variants. The prompt explicitly asks the model to refer to the user’s transcript and provide actionable suggestions. This design is necessary because LLMs can otherwise produce generic feedback even when retrieved context is available.

The no-retrieval baseline uses the same prompt structure but does not include retrieved documents. This allows us to isolate the contribution of retrieval. If retrieval-based methods produce more specific or actionable feedback than the baseline, the improvement can be attributed to the additional coaching context.

3.7 System Variants

We compare the following system variants:

- (1) **No Retrieval Baseline:** The LLM generates feedback using only the user transcript and scenario.
- (2) **BM25 Retrieval:** The system retrieves top- k corpus items using BM25 and passes them to the LLM.
- (3) **Dense Retrieval:** The system retrieves top- k corpus items using sentence embedding similarity.
- (4) **Tag-Filtered BM25 Retrieval:** The system first filters or re-ranks documents using tag overlap, then applies BM25.
- (5) **Tag-Filtered Dense Retrieval:** The system first filters or re-ranks documents using tag overlap, then applies dense retrieval.

This comparison is designed to answer three research questions. First, does retrieval improve feedback compared with a no-retrieval LLM baseline? Second, does dense retrieval provide better coaching context than sparse retrieval? Third, does tag-aware retrieval improve the relevance and usefulness of retrieved documents compared with retrieval based only on text similarity?

3.8 Implementation Challenges

Several challenges arose during the design and implementation of the system. The first challenge was corpus construction. The corpus needed to support both retrieval and analysis, so each item had to include not only text but also metadata such as scenario, document type, and tags. We solved this by using a structured JSON format containing a list of coaching documents. This format is easy to inspect, edit, and extend, while still being simple to load into the retrieval pipeline.

The second challenge was designing the tag set. A very detailed taxonomy could describe communication behavior more precisely, but it would also be harder to annotate consistently and would make retrieval more sparse. We solved this by using a compact tag set that captures common coaching strengths and weaknesses.

The third challenge was balancing semantic similarity with coaching relevance. Standard BM25 and dense retrieval may retrieve textually similar documents that do not address the user’s actual weakness. We addressed this by incorporating tag overlap into retrieval. Tags help the system retrieve documents that match the user’s coaching needs.

The final challenge was preventing generic LLM feedback. Even with retrieval, an LLM may generate broad suggestions such as “be more specific” without explaining how. To address this, the generation prompt requires the model to refer directly to the transcript, use detected tags, and provide concrete revision suggestions. This prompt design encourages more grounded and actionable feedback.

4 Experiments

In this section, we empirically evaluate our retrieval-augmented dialogue coaching system. We first describe the experimental setup and the five system variants we compare (Section 4.1), the dataset and corpus used in our experiments (Section 4.2), the five evaluation metrics that capture different qualities of coaching feedback (Section 4.3), and our LLM-as-judge evaluation framework along with the resulting quantitative and per-input analyses (Section 4.4). Together, these experiments are designed to answer whether retrieval improves LLM-generated coaching feedback, whether dense retrieval offers benefits over sparse retrieval, and whether tag-aware retrieval provides additional gains over standard text-similarity-based retrieval.

4.1 Experimental Setup

The goal of our experiments is to evaluate whether retrieval-augmented generation improves the quality of LLM-generated dialogue coaching feedback, and whether tag-aware retrieval provides additional benefits over standard text-similarity-based retrieval. We compare several feedback generation variants using the same input transcripts, corpus, and output format. This allows us to isolate the effect of retrieval strategy on the generated feedback.

Each system variant receives a user transcript and a scenario label, such as `elevator_pitch` or `behavioral_interview`. For retrieval-based variants, the system retrieves relevant coaching documents from the corpus before generating feedback. For the no-retrieval baseline, the LLM generates feedback using only the user transcript and scenario. All variants use the same feedback prompt structure, which asks the model to provide an overall assessment, strengths, areas for improvement, specific suggestions, and an improved version of the response.

We evaluate the following system variants:

- (1) **No Retrieval Baseline:** The LLM generates feedback without external retrieved context.
- (2) **BM25 Retrieval:** The system retrieves coaching documents using sparse keyword-based retrieval.
- (3) **Dense Retrieval:** The system retrieves coaching documents using sentence embedding similarity.
- (4) **Tag-Filtered BM25 Retrieval:** The system uses predicted coaching tags to filter or re-rank documents before applying BM25.
- (5) **Tag-Filtered Dense Retrieval:** The system uses predicted coaching tags to filter or re-rank documents before applying dense retrieval.

These variants are designed to answer three experimental questions. First, does retrieval improve feedback compared with a standalone LLM? Second, does dense retrieval retrieve more useful coaching context than sparse retrieval? Third, does tag-aware retrieval improve the relevance and usefulness of retrieved documents compared with retrieval based only on text similarity?

4.2 Dataset

We construct a small dialogue coaching dataset consisting of user transcripts and a retrieval corpus. The input set contains transcripts from communication coaching scenarios, including elevator pitches and behavioral interview answers. Each transcript is paired with a scenario label and a set of coaching tags that describe its main strengths and weaknesses.

The retrieval corpus is stored as a JSON file containing a list of coaching documents. Each document includes an identifier, document type, scenario, text content, and tags. The corpus contains three document types: `revision_example`, `feedback_example`, and `guideline`. Revision examples provide sample responses and improved responses, feedback examples provide critiques of sample responses, and guidelines provide general coaching principles.

We use a manually constructed dataset because the purpose of this project is to study retrieval strategies for coaching feedback rather than to build a large-scale benchmark. Manual construction also allows us to control the tag taxonomy and ensure that the corpus contains examples relevant to the selected coaching scenarios.

4.3 Evaluation Metrics

We evaluate generated feedback using five criteria: relevance, specificity, actionability, helpfulness, and groundedness. These metrics are designed to capture the qualities that make dialogue coaching feedback useful to a learner.

- (1) **Relevance:** Whether the feedback addresses the user’s actual transcript and scenario.
- (2) **Specificity:** Whether the feedback includes concrete observations rather than generic comments.
- (3) **Actionability:** Whether the feedback gives clear steps that the user can apply to improve the response.
- (4) **Helpfulness:** The overall usefulness of the feedback from a coaching perspective.
- (5) **Groundedness:** Whether the feedback is supported by the user’s transcript and, for retrieval-based methods, by the retrieved coaching context.

4.4 LLM-as-Judge Evaluation

Evaluating dialogue coaching feedback is challenging because there is no single correct output. A useful feedback response should be relevant to the user’s transcript, specific, actionable, helpful, and grounded in the provided input and retrieved context. Instead of relying on reference-based automatic metrics, we use an LLM-as-judge evaluation framework to compare feedback generated by different system variants.

For each input transcript, we generate feedback using all system variants. The resulting feedback outputs are then evaluated by a separate LLM judge. The judge receives the original user transcript, the scenario, the generated feedback, and, when applicable, the retrieved context used by the system. The judge assigns a score from 1 to 5 for each evaluation dimension: relevance, specificity, actionability, helpfulness, and groundedness.

We use LLM-as-judge evaluation because manual evaluation is time-consuming and difficult to scale across multiple system variants. LLM-based evaluation provides a consistent way to compare outputs using the same rubric. To reduce bias, the judge evaluates one output at a time using a fixed rubric rather than directly comparing named systems. The system name is not shown to the judge during scoring. We are attaching the judge prompt in Appendix A.

For each system variant, we compute the average score for each dimension across all test inputs. We also compute an overall average by averaging the five metric scores. The final LLM-as-judge evaluation results are reported in Table 1.

Table 1 shows several patterns. First, all five variants score in a narrow band (3.33–4.06), indicating that none of the methods produces dramatically poor feedback under the judge’s rubric. Second, the clearest effect of retrieval appears in groundedness: every retrieval-based variant scores higher (3.83–3.94) than the no-retrieval baseline (3.56), which is consistent with the intuition that retrieved coaching context gives the generator concrete material to anchor its claims. Third, Tag-BM25 obtains the highest actionability (4.06) and groundedness (3.94) scores, suggesting that tag filtering pulls in documents that prompt the model toward more concrete revision suggestions. Somewhat surprisingly, the no-retrieval baseline obtains the highest specificity (3.94) and relevance (3.56). One plausible explanation is that without retrieved context, the model focuses entirely on the user’s transcript and is not pulled toward generic phrasings from retrieved guidelines.

While the mean scores in Table 1 are close across methods, Table 2 shows that the per-input picture is sharper. When each input is attributed to the variant with the highest overall score, tag-aware

methods win the majority of inputs: Tag-Dense is the top method on 37.5% of inputs and Tag-BM25 on 31.3%, for a combined 68.8%. The two retrieval methods without tag filtering each win only a single input (6.3%), and the no-retrieval baseline wins 18.8%. This contrast between the aggregate and per-input views suggests that tag-aware retrieval is rarely dramatically better on any single dimension but is more often the best overall choice for a given transcript, which is exactly the behavior we would expect if tag filtering provides a small but consistent advantage in selecting pedagogically relevant context. The fact that the no-retrieval baseline still wins more inputs than plain BM25 or Dense also suggests that retrieval without tag filtering can sometimes introduce distracting context that hurts feedback quality, reinforcing the case for tag-aware filtering rather than retrieval alone.

Although LLM-as-judge evaluation is not a perfect substitute for human evaluation, it is appropriate for this project because it allows us to compare multiple generation methods using a consistent rubric. We treat the judge scores as approximate indicators of feedback quality rather than absolute ground truth.

5 Discussion

Our experiments were designed to answer three questions: whether retrieval improves feedback over a standalone LLM, whether dense retrieval outperforms sparse retrieval, and whether tag-aware retrieval provides additional benefits over text-similarity-based retrieval. The results give a nuanced answer to each.

Retrieval helps, but mainly through groundedness. The most consistent effect of adding retrieval is an improvement in groundedness, where all four retrieval variants (3.83–3.94) outperform the no-retrieval baseline (3.56). This is the dimension where retrieved coaching context is most directly useful: when the generator has access to feedback examples and guidelines, it produces claims that can be traced to concrete source material rather than the model’s internal priors. In contrast, retrieval does not uniformly improve specificity or relevance, and on these dimensions the no-retrieval baseline is actually competitive. We interpret this as evidence that retrieval changes what kind of feedback the model produces (more grounded, more actionable) rather than simply making it uniformly better.

Dense retrieval does not clearly beat sparse retrieval. We did not find evidence that dense retrieval consistently outperforms BM25 in this setting. BM25 matches or exceeds Dense on four of five dimensions, and the two methods tie on groundedness. Our corpus is small and manually constructed, so lexical overlap between transcripts and relevant documents is already high, which favors BM25.

Tag-aware retrieval helps most when measured at the input level. The aggregate scores in Table 1 understate the benefit of tag filtering: Tag-BM25 and Tag-Dense are rarely the single best variant on any individual dimension, but Table 2 shows that they are the top overall method on 68.8% of inputs, compared to 18.8% for the no-retrieval baseline and 12.6% combined for plain BM25 and Dense. This gap between the mean-score view and the per-input view is informative. It suggests that tag filtering does not produce a large effect on any one dimension but provides a small, consistent advantage across

Table 1: LLM-as-judge evaluation results on a 1–5 scale. Bold indicates the best score per column; underlined indicates the second-best. Values shown as mean $\pm$ standard deviation.

Method	Spec.	Act.	Rel.	Help.	Ground.
No Retrieval	3.94 $\pm$ 0.24	3.89 $\pm$ 0.47	3.56 $\pm$ 0.62	3.61 $\pm$ 0.61	3.56 $\pm$ 0.70
BM25	<u>3.72</u> $\pm$ 0.46	3.94 $\pm$ 0.24	<u>3.50</u> $\pm$ 0.51	3.72 $\pm$ 0.46	<u>3.89</u> $\pm$ 0.32
Dense	3.56 $\pm$ 0.51	<u>4.00</u> $\pm$ 0.34	3.39 $\pm$ 0.50	3.50 $\pm$ 0.51	<u>3.89</u> $\pm$ 0.32
Tag-BM25	<u>3.72</u> $\pm$ 0.46	4.06 $\pm$ 0.24	3.39 $\pm$ 0.61	<u>3.67</u> $\pm$ 0.59	3.94 $\pm$ 0.42
Tag-Dense	<u>3.72</u> $\pm$ 0.46	3.89 $\pm$ 0.32	3.33 $\pm$ 0.49	3.61 $\pm$ 0.50	3.83 $\pm$ 0.38

Table 2: Per-input winners under LLM-as-judge evaluation. Each input is attributed to the method with the highest overall score. Of 18 total inputs, 2 were tied at the top and are excluded; percentages are over the remaining 16. Bold = best, underlined = second-best.

Method	# Won	% of Inputs
No Retrieval	3	18.8%
BM25	1	6.3%
Dense	1	6.3%
Tag-BM25	<u>5</u>	<u>31.3%</u>
Tag-Dense	6	37.5%

dimensions that adds up to a better overall feedback response for most transcripts.

Why does the no-retrieval baseline still win on some dimensions?

A surprising finding is that the no-retrieval baseline obtains the highest specificity (3.94) and relevance (3.56) scores. We see two likely explanations. First, when no documents are retrieved, the model focuses entirely on the user’s transcript and is not pulled toward phrasings drawn from generic guidelines. This can produce feedback that reads as more directly tied to the specific transcript, even if it is less grounded in external coaching principles. Second, retrieval can occasionally introduce documents that are tangentially related, which may dilute the feedback’s focus. This points to a practical tradeoff: retrieval improves groundedness and actionability but can mildly reduce surface-level specificity unless the retrieved documents are well-matched to the user’s needs, which is precisely the gap that tag filtering is designed to close.

6 Conclusion

This work studied whether retrieval-augmented generation can improve the quality of LLM-based dialogue coaching feedback, and whether tag-aware retrieval provides additional benefits over standard text-similarity-based retrieval. We formulated dialogue coaching as a RAG problem, designed a compact tag taxonomy for representing communication strengths and weaknesses, implemented a tag-filtered retrieval pipeline combining coaching tags with sparse and dense retrieval, and evaluated five system variants using an LLM-as-judge framework across five dimensions.

Our experiments show that retrieval most clearly improves the groundedness of generated feedback, with all retrieval-based variants outperforming the no-retrieval baseline on this dimension. We did not find consistent evidence that dense retrieval outperforms BM25 in our small-corpus setting. Tag-aware retrieval provides a

small but consistent advantage: although Tag-BM25 and Tag-Dense rarely dominate any single evaluation dimension, they are the best overall method on 68.8% of inputs, compared with 18.8% for the no-retrieval baseline. Together, these results suggest that for subjective, pedagogical generation tasks, the relevance of retrieved documents matters more than the choice of retrieval algorithm, and that an intermediate tag representation is a useful way to bridge user diagnosis and document selection. We see this as a promising direction for building more grounded and actionable LLM-based coaching systems, and we discuss extensions briefly in Section 7.

7 Limitations and Future Work

Our evaluation uses 18 inputs and a small, manually constructed corpus, so per-dimension differences within the reported standard deviations are suggestive rather than conclusive. Tags are assigned manually, meaning our tag-aware results approximate an upper bound on realistic tag prediction. We also rely on a single LLM judge from the same model family as the generator, which may bias absolute scores. Future work should scale the dataset and corpus, replace manual tagging with an automated tagger, and validate the LLM judge against human ratings.

8 Contributions

Both authors contributed equally to this project. Grace led the initial codebase scaffolding, designed the LLM-as-judge rubric and prompt, and finalized the written report. Supia led the related work survey, implemented the retrieval and tag-filtering pipelines, developed the feedback generation modules, and finalized the codebase.

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A Prompt

The judge prompt follows the structure below:

You are an impartial evaluator of dialogue coaching feedback.

Scenario:

{scenario}

User transcript:

{transcript}

Retrieved context, if any:

{retrieved_context}

Generated feedback:

{feedback}

Rate the generated feedback from 1 to 5 on each criterion:

1. Relevance
2. Specificity
3. Actionability
4. Helpfulness
5. Groundedness

Definitions:

- Relevance: Does the feedback address the user's transcript and scenario?
- Specificity: Does the feedback mention concrete details from the transcript?
- Actionability: Does the feedback give clear steps for improvement?
- Helpfulness: Would this feedback be useful to someone practicing?
- Groundedness: Are the claims supported by the transcript and retrieved context?

Return your answer as JSON:

```
{
  "relevance": ...,
  "specificity": ...,
  "actionability": ...,
  "helpfulness": ...,
  "groundedness": ...,
  "brief_rationale": "..."
}
```

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